

Scania APS failure classifier

Datative GDGoC '26 · Binary root-cause triage for trucks that have already failed

94.9%	49.2%	0.882	93.0%
Test recall	Test precision	PR-AUC	Cost reduction

Key insight

APS failures are rare: 1,000 of 60,000 training cases (1.67%). Missingness is not random: six of the ten strongest model inputs are missing-value indicators. For br_000, missingness is 83.3% for non-APS failures versus 9.0% for APS failures, suggesting that measurement availability itself captures operating or diagnostic context.

Model and feature engineering

A balanced 320-tree Extra Trees classifier captures nonlinear sensor interactions. Median imputation is fitted inside the pipeline and missingness indicators convert the data-collection pattern into explicit features. All 170 anonymized inputs are retained; embedded random-subspace selection and importance analysis limit reliance on weak variables without imposing unsupported mechanical assumptions.

Threshold rationale

The decision threshold is 0.143, selected only on a stratified validation fold to minimize $10 \times \text{FP} + 500 \times \text{FN}$. On the untouched test set it produced 356 true APS detections, 19 missed APS failures, and 367 false APS alerts. Cost fell from 187,500 to 13,170 units.

EDA and data quality

60,000 training rows · 16,000 test rows · 170 features · 8.3% missing cells · 8 features over 50% missing · 0 exact duplicate training rows. A sampled train/test drift screen found a maximum KS statistic of 0.027, indicating close supplied-split alignment.

Business translation

The model is a service-center triage aid: it prioritizes trucks for APS inspection while accepting more false APS alerts to avoid expensive missed APS diagnoses. Neg means another system caused the failure; it does not mean the truck is healthy. Replace the published proxy costs with real inspection, downtime, and towing economics before production use.

Limitations and next step

Anonymized features prevent mechanical root-cause interpretation. No truck IDs or timestamps are supplied, so entity-level leakage and temporal stability cannot be tested. Validate prospectively on later trucks, audit whether diagnostic procedures create the missingness signal, and monitor missingness, alert volume, recall, precision, and realized cost.

Final model: median imputation + missing indicators + balanced Extra Trees. Test ROC-AUC: 0.994 · F1: 0.648.